L8.2: What is an orthonormal basis?

Transcript Language

English

Hello, and welcome to the Maths 2 component of the online B.Sc. program on data science

and programming. In this video, we are going to talk about what is an orthonormal basis.

So, let us just recall first that we have defined what is an orthogonal set.

So, an orthogonal set of vectors v1, v2, vk in an inner product space is a set of vectors

whose elements are mutually orthogonal. That means, if you take the inner product vi,vj,

where i is not equal to j, then this is 0. So, we checked in the previous video that an

orthogonal set of vectors is linearly independent. And that is why if you take a maximal orthogonal

set of vectors, then it is a maximal linearly independent set. And we have seen before that a

maximal linearly independent set is a basis. So, this is one way of getting a basis for an inner

product space. So, this is somewhat special. If you have an inner product on your vector space,

this is a slightly enhanced way of getting a basis. So,

such a basis was called an orthogonal basis. So, what is an orthonormal set? So, we are going

to use these two terms, an orthonormal set and a basis and put them together

and we will get an orthogonal, orthonormal basis. So, an orthonormal set of vectors

in an inner product space is an orthogonal set of vectors. So, that means the inner product of

vi and vj is 0 for all i not equal to j, such that the norm of each vector of the set is 1.

So, let is recall that, if you have an inner product, it automatically gives you a norm that is

if you have a vector v and you take the inner product of v,v, then that is a positive number,

non-negative number and if you take its square root, that gives you the norm. So,

that is defined as the norm of the vector. So, explicitly, what this means is if you have

a set v1, v2, vk then S is an orthonormal set of vectors if the inner product of vi and vj is

0 for i and j in 1 through k and i is not equal to j, and the norm of vi is 1. So, just to explicitly

say that norm of vi is 1 is the same as saying that inner product of vi,vi is equal to 1.

So, here, of course, we do not take square root because a square root of 1 is 1. So,

we know that if the norm is 1, this is the same as saying the inner product is 1.

So, as an example, let us consider R4 with the usual inner product, that is the dot product. So,

then the set minus 1 by root 3, 1 by root 3, 1 by root 3, 0, 2 by root 42, 1 by root 42, 1 by root

42, 6 by root 42 and two-third, 0, two-thirds, minus one-third is an orthonormal set of vectors.

So, this example, we have sort of seen a similar example before.

And the idea is here we have made them all into a norm, we have made all of them

to have norm 1. So, if you take the norm, so norm of the first vector is,

so this is the usual inner product, so the norm is just given by taking each component, squaring it,

and adding it up. So, that gives us one-third plus one-third plus one-third, which is 1.

So, similarly here, if you take the norm, that is going to give you 4 by 42 plus

1 by 42 plus 1 by 42 plus 36 by 42, which is indeed 1. And similarly, over here,

the norm is going to be given by two-third square plus two-third square plus minus one-third square,

that is 4 by 9 plus 4 by 9 plus 1 by 9, which is 1. So, all these have norm 1. And

you can check that the inner product is 0 if you take two different vectors.

So, for that, since all of them have the same denominator, you can ignore the denominator and

take the inner product and you can see clearly that the norm is indeed 0, sorry, the inner

product is indeed 0. So, I hope it is clear what is an orthonormal set. It is just an orthogonal

set with the additional property that the norms of each of the vectors in that set is 1.

So, now, what is an orthonormal basis? So, we know what is an orthonormal set. So, now,

an orthonormal set which forms a basis is an orthonormal basis. So,

this is similar to what we saw for the orthogonal basis, namely, an orthogonal basis was one where

it was an orthogonal set and it was a basis. So, here an orthonormal basis is one where it is

a basis and it is an orthonormal set.

So, equivalently an orthonormal basis is an orthogonal basis where the norm of each

vector is 1. So, because an orthogonal set is, orthonormal set is nothing but

an orthogonal set where each vector has norm 1 and orthonormal basis in particular is an orthogonal

basis where each vector has norm 1. So, equivalently, an orthonormal basis is

a maximal orthonormal set. So, just to make it clear what we mean by that, that means this set

is an orthonormal set and there is no set which is bigger than this, which properly contains this set

and which is also an orthonormal set. So, this is the largest possible orthonormal set you can

get. You cannot expand this set further and retain the property of being orthonormal.

So, here is an example. The standard basis with respect to the usual inner product

forms an orthonormal basis, maybe let us check that quickly before going ahead.

So, I have the vectors ei. So, we already know that if you take ei,ej, then this is

and i is not equal to j, then this inner product is going to be,

so if you work this out, 0 times 0 plus 0 times 0 all the way, so, there is a lot of 0s.

And then when you come to the ith place, you will get a 0, sorry, a 1 time 0.

And when you come to the jth place,

you will get 0 times 1, this is if i is larger than, sorry, less than j then you get this. If

i is larger than j then the 0 times 1 comes first and the 1 times 0 comes next,

either way the point is you get this to be 0. So, this inner product is 0 and the norm of ei,

so that is ei,ei, so, well, square root of this, so then if you do the same computation,

you get 0 times 0 plus bunch of 0s. And then in the ith place there is one in each component,

so 1 times 1 and then again 0s. So, this gives you, I should have square root of this,

so which is square root of 1. And of course, when we say square

root and we are talking about norms, we always take the positive square root,

so, this is again 1. So, this is an example of an orthonormal basis. It is an orthogonal basis.

Well, we already know it is a basis. It is an orthogonal set that is what we checked

over here. So, here we checked it is orthogonal and here we checked it is orthonormal.

Let us do another example. So, consider R3 with the usual inner product and the set

one-third 1, 2, 2, one-third minus 2, minus 1, 2 and one-third 2, minus 2, 1,

then this set which we have called beta is an orthonormal basis. So, let us check this.

So, let us look at the norms first. So, let us give these names. So, let us call this,

let us call the first vector v1. So, we have v1, v2 and v3. So, the norm of v1 is

square root of, so, typically, if you want to, if you want to check that something has norm 1, then

it is enough to check that its square is 1. So, instead of checking that norm v1 is 1, I will

be checking norm of v1 squared is 1. So, norm of v1 squared is just the inner product of v1 itself,

which we can compute easily. So, this is one-third 1, 2, 2,

one-third 1, 2, 2, and that is 1 by 9 times 1 times 1 plus 2 times 2 plus 2 times 2 which is

1 plus 4 plus 4 so 9 by 9 which is 1. The same, the same thing is going to happen

if you take v2, norm of v2 which is one-third minus 2, minus 1, 2, one-third minus 2, minus 1,

2. So, this is giving us 1 by 9 times minus 2 times minus 2 plus minus 1 times minus 1

plus 2 times 2, which is 9 by 9, which is 1. Maybe I will leave norm v3 square

to you. So, check that this is 1. And then we are left with these three

inner products. So, let us check v1, v2 what is the inner product. So, we have one-third 1, 2, 2,

and one-third minus 2, minus 1, 2. So, again, the tip here is, if you want to check that the inner

product is 0, then these constants you can remove out and check for the term inside, which if it

is better term, so in this case, for example, you have integers,

then you would rather check that. So, in this case, you get 1 by 9 times

1 times minus 2 plus 2 times minus 1 plus 2 times 2. So, that gives us minus 2 minus 2 plus 4. So,

that is 0. I will again encourage you to check the other terms. So, this shows that this is

an orthonormal basis. Why is it a basis? We have checked here that this is an orthonormal

set, but the reason it is a basis is because it is a maximal orthonormal set.

Meaning, this is a set of size 3 and you already know it is a linearly independent set, because

if it is an orthonormal set, it is in particular an orthogonal set, which we have checked is

linearly independent. So, this is a, so I should end this by saying once you finish this checking,

this will so, this part will show and along with v3 will show that the norms are 1.

This part if you finish, so check also that v1, v3 is v2, v3

is 0, this will, altogether will show that it is orthogonal. And then

the cardinality of beta is 3 and beta is linearly independent,

because it is orthogonal and we know that dimension of R3 is 3.

So, that implies that beta is an orthonormal basis,

so because the sizes match. So, you have a linearly independent set of size which

is equal to the dimension of the vector space that is why it is a basis. So, in particular,

it is an orthonormal basis in this case. So, we have, I hope this gives you a window

into how to check something is a basis. Sometimes you happen to not know an explicit basis, but you

know the dimension of the space in consideration, so you can check that something is a basis.

So, let us talk about obtaining orthonormal sets from orthogonal sets.

So, suppose you have inner product space, an inner product space and if gamma is a set v1, v2, vk,

which is orthogonal, so it is an orthogonal set of vectors. So, then we can obtain an orthonormal

set of vectors which let us call it beta from gamma by the following procedure. You take each

of these vectors and divide it by its norm then we are claiming that this is an orthonormal set.

So, why is it an orthonormal set? So, the reason is because this is exactly what we did in the

previous example. Each of those vectors was divided by its norm which made it orthonormal

and the orthogonality was evident already from just the integer part. So, here also that is

what is happening. So, here we have that v1, v2 is, so let us say vivj is 0, so this will imply

that vi by norm vi, vj by norm vj is 0. And so, this is orthogonal. So, this new

set is orthogonal beta. So, the only thing that is remaining to check

is what is a norm. So, if you take the norm of vi by norm vi, well, constants come out of the norm.

So, you get 1 by norm vi times norm vi, which is 1. So, that is why this is an orthonormal set.

So, let us consider R2 with the usual inner product and the orthogonal basis

gamma one-third, minus 3, 1. So, we can make this into an orthonormal basis by looking at

the norms of each of these vectors and then dividing by the term. So, the norm of both of

these vectors turns out to be root 10. So, if you do 1 by root 10 times 1, 3 and then 1 by root 10,

1 by root 10 times minus 3, 1 this is an orthonormal basis for R2.

So, the fact that it is an orthogonal set is already because gamma was orthogonal. The fact

that it is orthonormal set is because we divided by the norms. And the fact that it is a basis,

well, gamma was already a basis. So, from there it follows this is a basis or the fact that R2 is of

dimension 2 and this is a orthonormal set of size 2 so it must be linearly independent, so linearly

independent of size 2, which tells you it is a basis.

So, maybe let me just write that again here, since it got lost out. So, here I was just saying that

vi,vj is 0 implies vi, norm vi, vj divided by norm vj. So, this is 1 by norm vi times norm vj times

the inner product of vi,vj, which is 0. So, from here, we get that the existing set is orthogonal,

sorry, the new set is orthogonal. And norm of vi by norm vi

we can take the norm out. So, this gives us 1.

So, that is what, that is why this statement holds true. And the example

tells you, if you do not understand what I have written here, tells you why these things work. So,

work out the example for yourself. So, why are orthonormal bases important?

This is a sort of the punch line of what we are doing. So, suppose gamma is v1, v2, vn and this is

an orthonormal basis of an inner product space V and suppose you have a vector v inside capital V.

Well, we already know because it is a basis that you can write little v as c1v1 plus c2v2

plus cnvn. Why is that, because remember that a basis is in particular a spanning set, a basis

is a spanning set, which means every vector is a linear combination of the basis vectors.

And in fact, for a basis, it is a unique linear combination. So, there are a unique c1, c2, cn

such that little v c1v1 plus c2v2 plus cnvn. So, this is a general statement for any basis. Now,

what is the importance of it, being an inner product space and with an orthonormal basis?

So, how do we find c1, c2, cn, for any basis this means writing

a system of linear equations and solving. So, that is how I typically solve for c1, c2,

cn. You write your v and then you write your v1, v2, vn and then you solve these equations.

But since gamma is orthonormal, we can use the inner product and compute ci is v,

inner product of v, vi. So, how do I do that? Let me quickly show you why that is true.

So, let us compute what is inner product of v,vi. So, if you compute what is inner product of v,vi,

I substitute c1v1 plus c2v2 plus civi plus cnvn and then inner product with respect with vi, well,

we know that the inner product is a linear in each variable, which means that I can write this as

c1 times inner product of v1,vi plus c2 times inner product of v1 v2,vi, all the way up to,

then we have plus ci times inner product of vi,vi plus

cn inner product of vn,vi. This is what we get.

But now, we know that this is an orthonormal basis. So, because it is an orthonormal basis,

first of all, it is orthogonal. So, other than the vi,vi term, every other term is going to be 0. So,

this term will remain and all these terms are 0. So, this is, I can just write this as ci times

vi,vi. So, this, we have used here that it is orthogonal, but now we also know

it is orthonormal, because it is orthonormal this is, we can rewrite this as ci times norm

vi squared, and norm vi we know is 1. So, this is just ci. This is where we are

using the fact that we have an orthonormal basis. So, what did we get? We got that

the inner product of v and vi is equal to ci. So, this is a very easy way of saying what

are the coefficients which come into this linear combination, which gives you v. How do we find c1,

c2, cn, the answer is each ci is take the inner product of v with vi and that is your ci.

Let us do an example. We saw this orthonormal basis for R2 earlier.

So, write 2,5 as a linear combination in terms of these basis vectors. So, if you take 2,5

let us write this as c1 times 1 by root 10 1,3 plus c2 times 1 by root 10 minus 3,1.

Well, we just saw, what is c1. c1 is the inner product of v,v1.

So, here v1 is 1 by root 10 times 1,3, which is, you can pull out the constant, so 1 by root 10

and then the remaining things, take the inner product, so this is 2 times 1 plus 5 times 3.

So, what does that give us, 1 by root 10 times

2 plus 15, so that is 17. And then we get c2, which is v,v2,

so 1 by root 10 times minus 3,1 which gives us,

again, the 1 by root 10 comes out, and we get 2 times minus 3 plus 5 times 1, which is minus

6 plus 5, so minus 1 by root 10. So, this is what we obtained as the coefficients.

So, now, if we write 2,5, in terms of these vectors we get 2,5 is 17 by root 10 times v1

plus minus 1 by root 10 times v2, so which if you write out completely 17 times root 10 times

1 by root 10 times one-third, sorry, 1,3 plus or rather minus 1 by root 10 times,

1 by root 10times minus 3,1. And you can check actually that this is indeed going to work.

If you want, we can do that quickly. This is 17 by 10 1,3 minus

1 by 10 minus 3,1. And if you work this out, the first entry is 17 by 10 minus minus 3 by 10. So,

17 by 10 plus 3 by 10 so 20 by 10, which is giving you 2. And then the second entry is 51 by 10 minus

1 by 10, which is giving you 5. So, this shows that, indeed, what the equation we wrote down is

correct. This would have been maybe slightly harder if we did not have this knowledge that

this is an orthonormal basis. So, I hope in this video you have

figured out what, I mean, you have understood the main point. The main point is that we define

something called an orthonormal basis, that is an orthogonal basis in which each vector has

norm 1. And the crux of the video is that every vector which you can write as a linear

combination of the orthonormal basis, the coefficients of, in that linear combination

c1, c2, cn are essentially are equal to the inner product of the vector v with the corresponding

basis vector vi. That is the main

point. So, I guess that finishes this video. Thank you.